

Automated Certificate Lifecycle Management Solution

Executive Summary

Due to the reduction of certificate validity from 365 days to 45 days, manual certificate management is no longer operationally sustainable. The proposed solution introduces end-to-end automation for certificate request, issuance, deployment, validation, and retirement across the Cloud Proxy infrastructure.

This solution eliminates manual intervention, reduces outage risk, improves compliance, and enables scalable certificate lifecycle management.

Current State Challenges

Manual Process

1. Track certificate expiry dates.
2. Raise renewal request in Venafi.
3. Download newly issued certificate.
4. Deploy certificate to Cloud Proxy.
5. Restart or reload services.
6. Validate certificate installation.
7. Update inventory records.

Risks

- Human error during deployment.
 - Missed renewal windows.
 - Service outages caused by expired certificates.
 - Increased operational effort due to 45-day lifecycle.
 - Lack of centralized visibility.
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Proposed Automated Solution

Architecture Components

Certificate Authority Platform

- Venafi as certificate lifecycle manager.
- Policy-driven certificate issuance.
- API-based integration.

Automation Orchestrator

Examples:

- Jenkins
- GitLab CI/CD
- Ansible Automation Platform
- Rundeck
- Internal Automation Service

Responsibilities:

- Monitor expiry dates.
- Request certificate renewal.
- Retrieve certificates from Venafi API.
- Trigger deployment workflows.

Secret Storage

- HashiCorp Vault
- Cloud Secret Manager
- Encrypted Certificate Repository

Responsibilities:

- Secure certificate storage.
- Version management.
- Access control.

Cloud Proxy Layer

Examples:

- NGINX
- HAProxy
- F5
- Envoy
- Cloud-based reverse proxies

Responsibilities:

- Automated certificate deployment.
 - Configuration validation.
 - Graceful reload.
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Automated Workflow

Day 0

Certificate issued and deployed.

Day 30

Automation identifies certificate approaching renewal threshold.

Day 31

System automatically:

1. Calls Venafi API.
2. Requests new certificate.
3. Retrieves certificate package.
4. Stores certificate securely.

Day 32

Automated deployment:

1. Backup existing certificate.
2. Deploy new certificate.
3. Validate certificate chain.
4. Validate hostname bindings.
5. Validate expiration date.
6. Reload proxy service.

Day 32

Post-deployment verification:

- SSL handshake validation.
- Health checks.
- Synthetic traffic testing.

Day 32

Notification sent:

- Success notification.
 - Audit log generated.
 - CMDB updated.
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Failure Handling

Automatic Rollback

If validation fails:

1. Restore previous certificate.
2. Reload service.
3. Generate incident ticket.
4. Notify support team.

Escalation

- Email alert
 - Teams/Slack notification
 - ServiceNow ticket creation
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Security Controls

- API authentication using service accounts.
 - RBAC-controlled certificate access.
 - Certificate private keys never exposed to users.
 - Audit trail for every certificate action.
 - Automated compliance reporting.
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Benefits

Operational Benefits

- 90%+ reduction in manual effort.
- Elimination of certificate expiry incidents.
- Faster certificate deployment.

Security Benefits

- Short-lived certificate compliance.
- Improved cryptographic hygiene.
- Centralized governance.

Business Benefits

- Reduced outage risk.
 - Improved compliance posture.
 - Scalable management for future certificate policy changes.
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Future Enhancements

- ACME-based certificate issuance.
 - Self-service certificate onboarding.
 - Automated certificate inventory dashboard.
 - Predictive renewal analytics.
 - Multi-cloud certificate lifecycle management.
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Success Metrics

- Zero certificate-expiry incidents.
- 100% automated renewals.
- Less than 15 minutes deployment time.
- 100% audit trail coverage.
- Reduction in manual certificate-related change requests.